A Supporting text of the numerical code

This is a user-friendly description of the MATLAB code of a seismo- thermomechanical model (STM) that investigates in the development of earthquakes and their time series when there is a pre-existing normal fault in a homogeneous medium crosscutting a depleting gas reservoir. The code is subdivided in the main code and the supporting files. In the section 'Code description' the different MATLAB files and functions are discussed and in the 'Input' section the different input parameters are shown and where you can fill them in. In the 'Output' chapter the output of the code is shown and the different figures that can be generated with the code are shown. In the last subsection of this appendix the assumptions and the most important statements are stated, including the limitations of the current model.

This model investigates in the development of the earthquakes in the depleting gas reservoir of Groningen. The specific region that is taken for investigation in this model is the Zeerijp region in Groningen. The grid of this model is fully stagerred and can be seen in figure 1. The further model setup of subsurface with homogeneous rock properties (elastic and frictional) is shown in figure 2, in which the fault is also indicated in the figure. The top of the model is at a depth of 2000 meter and the bottom is at 4000 meter depth. the middle of the reservoir layer is placed at 2950 meter depth and has a thickness that is adjustable, but for the reference model is stated at 200 meter. The offset between the left and right side of the fault plane is 50 meters. In figure 3 the model is shown with a layered lithology of the Groningen subsurface. With this setup of the model there can be investigated in the relation between the different layers in the Groningen gasfield and their frictional and elastic properties and the earthquake nucleation in this area.

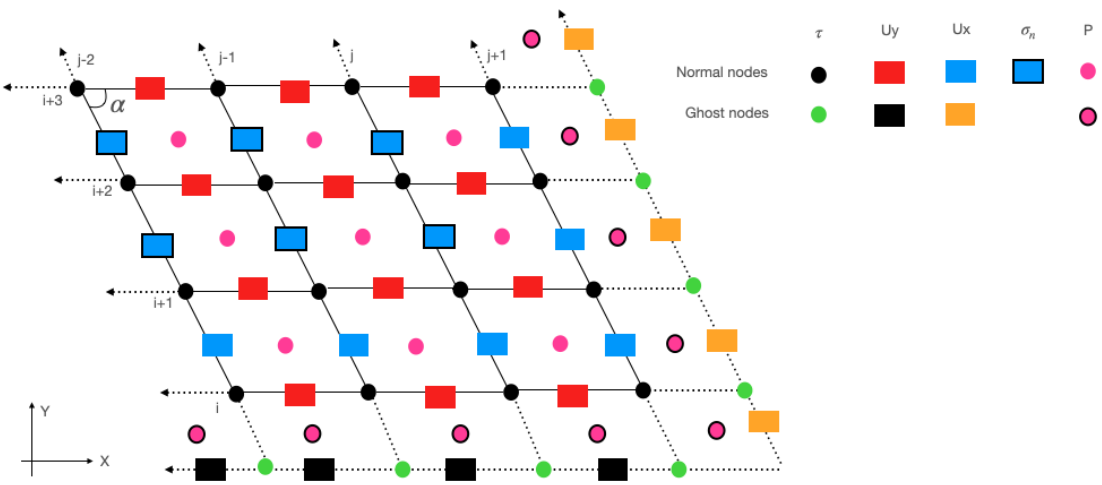

Figure 1: The indexing of the fully staggered grid and the corresponding gohst nodes of the model. The model domain is where the lines between the different nodes are solid. The indexes i and j correspond to the numbering of the grid lines of the model in vertical and horizontal direction. The model is tilted with the angle α.

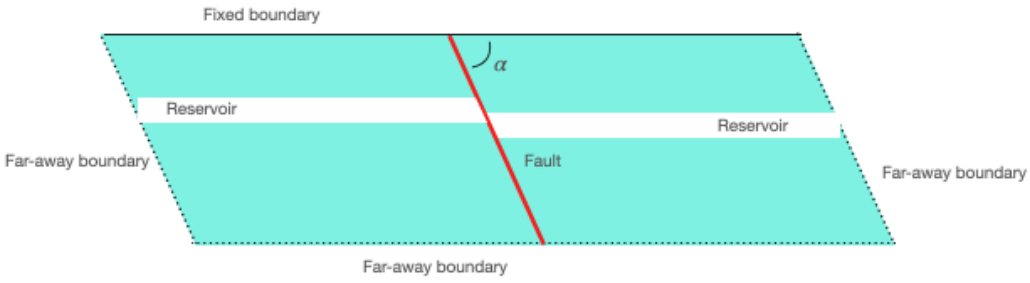

Figure 2: The setup of the 2D plain strain model. The upper boundary is represented as a solid line due to the fixed top boundary condition. The other three boundaries are modelled as far-away boundaries and are therefore represented with dotted lines.

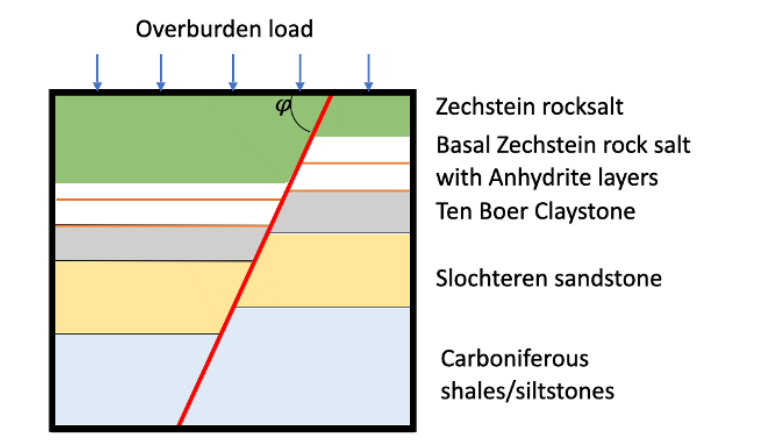

Figure 3: Setup of the seismothermomechanical model with the layered lithology of Groningen. The different layers are based on the research of Hunfeld et al. (2017) .

A.1 Code description

Main-code.m is the main file of the code. In this file all the calculations are done. If this file is runned, the code will generate an output. In this file the multiple different MATLAB functions that are represent in the ZIP folder are used. In the beginning of the code the input file is read in and there can be made different choices about the plotting style of the figures, the running time of the model and the interval at which all the different values will be stored. Also the input.mat file is loaded into the code. In this file all the input parameters are stated. This makes it relatively easy to change the different input statements for different investigations of the earthquake nucleation in Groningen. After the loading has been done, the numerical grid will be generated (as in figure 1). The next step there can be chosen for a time and a time interval step. The frictional parameters a and be can be changed in the lines after this. The code will also state the inital stresses in the subsurface (chosen by using the geological survey of Buijze et al. (2019), following the lithostatic pressure gradient with a hydro-static fluid and gas pressure) and then the code will start with building the right hand side and the left hand side of the PDE's that are stated in the method section of this thesis. The calculations that will be then done are:

- The rate-and-state solver.
- Addeptive time stepping.
- Aging law.
- Direct solver of the PDE's.
- Slip calculation on the fault plane.
- Stress calculation of the subsurface with strain and Hook's law.
- Pressure calculations.

The output is generated and saved in an file called 'output.txt'. Also, multiple plots and even a movie can be shown by running this code and the plots can also be saved in the last part of the code. For more information about the output of the code, please read the 'output' subchapter.

A.2 Input in the model

The input that can be given in the model is listed below and in the same order as in the generate-input.m file in the ZIP code.

- Dip of the fault plane (α) in degrees.
- Horizontal model size in meters.
- Vertical model size in meters.
- Number of nodes in the horizontal direction of the model.
- Number of nodes in the vertical direction of the model.
- Density of the rock (for homogeneous model setup) in kg/m^3 (ρ).
- Density of the fluid in the reservoir in kg/m^3 (ρ_f).
- Density of the gas in the reservoir in kg/m^3 (ρ_g).
- Far field loading rate in m/s (V_p).
- Shear wave speed in m/s (c_s).
- Poisson ratio of the rock (ν).
- Gravitational constant in m/s (g).
- Bulk modulus of the rock (K_0).
- Dynamic slip weakening velocity (V_w).
- Background slip rate (V_i).

- Initial friction coefficient (μ)
- Initial reference slip rate (V)
- Rate and state direct effect (a)
- Rate and state evolution effect (b).
- Characteristic slip distance (L).
- Pressure depleting rate (P).

A.3 Output of the model

The model can generate different figures to shown the earthquake nucleation of the Groningen gasfield. The main output of the file is that each time step is saved in a text file called 'output.txt'. Other outputs that are generated from the code are MATLAB files of the input of the code, the data at different time steps. This is saved for the user friendly advantages. Because, when saved, the used does not have to run the script again, but can just load in the saved data points. The output of the code at every time step is:

- The elapsed time in seconds of the model,
- The running time of the code itself.
- The maximal slip rate of the fault plane.
- The minimal slip rate of the fault plane.
- The maximal displacement.

With these outputs graphs can be made. (INCLUDE SOME GRAPHS AND WHERE TO MAKE THEM IN THE CODE).